

Genre & Lyrics: Using Language Models to Explore this Relationship

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Abstract

This study investigates the relationship between music genres and song lyrics, examining whether lyrics can reliably indicate a song's genre and how effectively genre-specific lyrics can be generated. We utilized transformer-based models, employing BERT for genre classification and GPT-2 for lyric generation, using a dataset of 2.94 million songs across various genres. Our approach integrates these tasks, evaluating the genre alignment of the generated lyrics through classification. The results indicate that while BERT achieves reasonable classification accuracy, genre boundaries often blur, particularly for overlapping styles such as pop and rock. The generated lyrics partially align with the intended genres, as confirmed by automated classification and human evaluation but do not fully capture their intended genre. These findings highlight the nuanced interplay between lyrics and genre.

1 Introduction

Music genres often evoke specific themes and ideas, suggesting that lyrics may reflect and define a song's genre. For example, country music typically features imagery related to rural life, such as trucks and open landscapes, while rap commonly incorporates urban narratives and often contains explicit language. This connection leads to an intriguing question: can a song's genre be determined solely by its lyrics? This study aims to explore this question through two interconnected tasks: classifying song genres based on lyrics and generating genre-specific lyrics. Existing research has primarily focused on genre classification and generating lyrics in isolation, while this study focuses on the intersection of the two.

The study employs two language models to achieve its objectives: BERT_cased for genre classification and GPT-2 for lyric generation. These models were trained and tested on a dataset of 2.94 million songs labeled by genre and sourced from

Hugging Face. Utilizing both models allows for a more in-depth exploration of the relationship between genre and lyrics. By combining BERT's classification capabilities with GPT-2's generative abilities, we can gain insights not only into how effectively lyrics can be classified by genre but also into how well the model can generate lyrics specific to each genre. This dual approach is implemented by using the classification model to evaluate the generated songs. If for instance, the classification model can accurately classify the generated songs, it indicates that both models have a certain understanding of genre and that lyrics can serve as an indicator of genre.

Our findings reveal a complex relationship between song lyrics and musical genres, showcasing both the potential and limitations of using language models for this analysis. The BERT_cased classification model demonstrated moderate success in identifying genres from lyrics; however, it tended to overpredict broader categories, such as pop. This highlights the challenges associated with genre overlap and imbalances in the dataset we worked with. Similarly, while the GPT-2 generation model was able to produce lyrics that somewhat aligned with genre-specific themes, the classification model struggled to consistently identify the intended genres, especially in the case of generated songs. These results indicate that although lyrics can serve as partial indicators of the genre of songs, overlapping themes across different genres create significant challenges for both classification and generation. This emphasizes the need for more advanced models and balanced datasets to capture the complex relationship between lyrics and musical genres more effectively.

2 Background

Understanding how song lyrics relate to musical genres is a challenging but intriguing problem at the intersection of natural language processing and

music. These questions build upon two primary assumptions: first, that lyrical content embodies patterns reflective of a song’s genre, and second, that genre-specific language can be learned and replicated by NLP models.

Conceptual Foundations Musical genres often define thematic, cultural, and structural elements in song lyrics. For instance, country music is associated with rural imagery and storytelling, while rap frequently features urban narratives and a distinctive use of rhythm and rhyme. Despite these patterns, genres are not rigidly defined, and overlaps occur frequently. Lyrics from pop songs, for example, may share thematic elements with rock or R&B, making genre identification a nuanced and complex task. Additionally, lyrics are not the only indication of genre, with many other elements contributing to what makes a genre a particular genre.

Methodological Foundations This research employs two primary methods: genre classification and lyric generation. For the classification task, we use BERT_cased, a transformer-based model, to identify the genre of songs. In the lyric generation phase, we utilize GPT-2, to attempt to produce contextually coherent and stylistically consistent text. The reason these models were chosen was time and resource constraints. While the performance of these tasks could be hypothetically improved with larger models, using them was not practical given our limitations in time and resources.

Addressing Gaps in the Literature While previous research has examined genre classification using song lyrics, there has been limited work that combines classification with lyric generation to assess genre alignment. This study addresses this gap by integrating these two tasks, offering an objective measure of genre adherence through classification models while also exploring how effectively generated lyrics reflect their intended genres. This dual approach allows for a more thorough analysis of the relationship between genre and lyrics.

3 Methods

This study explores the relationship between song lyrics and genre using a dual approach: a classification task to evaluate the alignment of lyrics with genres and a generation task to produce genre-specific lyrics. The classification model uses BERT_cased to analyze existing lyrics and identify

their genres, while the generation model, GPT-2, creates new lyrics prompted by genre and a starting lyric. The following subsections provide a detailed overview of the models, datasets, training process, and evaluation approach. All code details can be found here ([Tauber and Heakins](#)).

3.1 Models

Classification Model We employed BERT_cased, a transformer-based model for the classification task. BERT was chosen due to its strong performance on text classification tasks, its ability to capture contextual relationships in text, and its established success in NLP research([Face, b](#)). It was also chosen for its ability to be trained quickly and with minimal resource usage. This was necessary due to the scope and size of this dataset.

Generation Model For lyric generation, we initially utilized DistilGPT due to its ability to produce coherent and contextually relevant text while using fewer computational resources. However, because DistilGPT was trained quickly, we decided to employ GPT-2([Face, c](#)) instead. This choice aimed to leverage the larger model’s capacity, with the expectation that it would outperform DistilGPT. Consequently, DistilGPT was discarded.

3.2 Datasets

Dataset Overview The study used a dataset of approximately 2.94 million songs labeled by genre, sourced from Hugging Face([Shah](#)). Genres included pop, rap, rock, country, and a miscellaneous (misc) category. The dataset was split into training (80%), validation (10%), and testing (10%) subsets.

Dataset Preprocessing Before training, the dataset underwent preprocessing to ensure its suitability for both classification and generation tasks. This included:

- Standardizing formatting into tsv files with necessary columns.
- Ensuring prompts for the generation model were structured as: "Genre: Specified genre name. Starting Lyric: The initial line for the generation model."

Data Imbalance The dataset showed an imbalance in genre representation, with pop and rap being the most represented genres, and country and misc the least represented. This dataset imbalance

was not addressed in our methodology however it does inform our findings. Future research in this area would need to create a dataset of equal representation to remove this implication from the findings.

3.3 Training

3.3.1 Classification Model Training

The BERT_cased model was fine-tuned on the labeled dataset using the NLP Scholar Pipeline for model training as described here (Prasad and Davis, 2024). The main hyper-parameter of note for us was the number of epochs this model was trained. We found the best performance to be in our 5 epoch model, as the more trained models suffered from over-fitting.

3.3.2 Generation Model Training

The GPT-2 model was trained to generate lyrics using the structured prompts.

Fine-Tuning Process: Fine-tuning involved adapting GPT-2 to the specific task of generating genre-aligned lyrics using the Hugging Face Trainer API (Face, 2024). Key considerations included:

- **Batch Size and Gradient Accumulation:** A batch size of 8 per device was used, with gradient accumulation over 2 steps to simulate a larger effective batch size while accommodating hardware limitations.
- **Mixed Precision Training:** Enabled through the fp16 flag to improve memory efficiency and training speed on GPUs.
- **Evaluation During Training:** The eval_strategy="epoch" parameter ensured periodic evaluation on the validation set to monitor loss and model
- **Memory Allocation:** The hyper-parameters in this stage were selected to enhance our training speed within the hardware limitations we encountered, primarily that the generation model was trained using a single NVIDIA 3060 12GB.
- **Number of Epochs:** Four epochs were selected as the optimal training length. This decision was based on the observation that loss plateaued after this point, indicating no further improvements in model performance

(see the section *Evaluation During Training* for further details).

Hardware Utilization: The training process leveraged GPU acceleration using PyTorch and the Accelerate (Face, a) library, optimizing data and model parallelism. Due to the constraints we faced using Colgate's supercomputer we moved to local training for ease of training.

Evaluation During Training: The main metric monitored during training was the loss, which was assessed at the end of each epoch. This provided a quantitative measure of the model's ability to generate lyrics that align with specific genres. The training process was stopped after four epochs because the loss had plateaued suggesting that further training would likely not enhance the model's performance. Although additional custom metrics for coherence or genre adherence were not implemented during training, the periodic evaluation of loss proved to be a reliable criterion for determining when to stop training. Future work could investigate the incorporation of more nuanced metrics to assess the alignment between generated lyrics and their intended genres during training.

3.4 Evaluation of Classifier

To evaluate the classification model, we used a 10% split of the dataset for testing. This subset was set aside during training to ensure an unbiased assessment of the model's classification performance.

We employed several standard metrics for evaluation: accuracy, precision, recall, and F1-score. Accuracy served as our primary metric, but we also reported precision, recall, and F1-score to provide a more detailed understanding of the model's performance across different genres. This was particularly important in cases where the dataset might be imbalanced, or when certain genres were more challenging to classify.

Our evaluation included a confusion matrix to identify patterns in the model's misclassifications, helping us to see which genres were frequently confused with one another. These heat maps were especially useful in pinpointing specific pairs of genres, such as pop and rock, that tended to be misclassified due to overlapping lyrical themes.

The evaluation methods we used were appropriate, as they provided a comprehensive understanding of the model's genre classification capabilities, areas of difficulty, and potential reasons for misclassifications. Future work may explore incorporating

more advanced evaluation metrics or techniques to further refine genre classification, especially in situations where genre boundaries are less distinct.

3.5 Generation Process

The song generation process utilizes the previously described trained GPT-2 model.

3.5.1 Key Steps

- **Model and Tokenizer Setup:** The `AutoModelForCausalLM` is used to load the fine-tuned GPT-2 model, and the tokenizer is set to pad sequences to the left using the model's `eos_token` to ensure the model generates text.
- **Generating Lyrics:** For each prompt, the model generates lyrics by predicting the next tokens iteratively. Key parameters:
 - `max_new_tokens`: Limits the generation to 1024 tokens, minus the length of the prompt. This ensures the generated output stays within the model's token limits.
 - `do_sample`: Enables Sampling. This introduces variability in the model's outputs, allowing for more creative and diverse lyrics.
 - `repetition_penalty`: Set the value to 1.01 to discourage repetition in the generated text. This specific parameter was necessary because using no repetition penalty led to the model repeating the same starting lyric. Conversely, a penalty set too high resulted in songs that were only one line long.
- **Output Collection:** The generated text is decoded and saved to a file, with each song separated by a delimiter for ease of processing later: `\n*****SONG STARTS HERE*****\n`.

3.6 Human Benchmark for Genre Classification

To evaluate the performance of our classification model, we collected human benchmark data. Three annotators were tasked with classifying 50 songs from various genres (excluding MISC due to its challenging classification) across both the test dataset and the generated songs dataset. The combined results of their classifications provide a benchmark for comparing the accuracy of our model.

4 Results

4.1 Comparison of Classification Model vs. Human Performance on the Test Dataset

In this subsection, we compare the performance of the classification model to human performance on the test dataset. Our model significantly over-

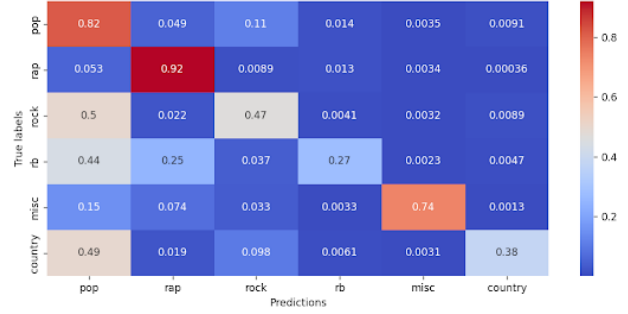


Figure 1: Heatmap of the classification model's performance on the test dataset.

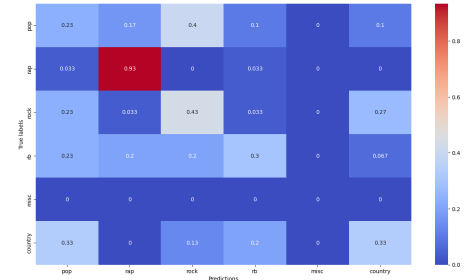


Figure 2: Heatmap of the aggregate human performance on genre classification for actual songs.

predicted the pop genre, as illustrated in Figure 1. Specifically, three genres—rock, R&B, and country—were misclassified as pop more often than they were correctly labeled. This discrepancy may be attributed to the skew in our dataset, where pop is the most represented genre. However, since rap is the second most represented genre, we would expect a similar misclassification for it. Instead, the over-classification of pop can be explained by the genre's vagueness, as it borrows elements from many other genres. This finding is further supported by our human-annotated results shown in Figure 2, which reveal a similar trend of misclassification, with an over-representation of pop for adjacent genres.

Moreover, the human baseline highlights the complexity of this task, as human annotators had a lower accuracy rate for every correct label except rap when compared to our model. Although

our model’s performance was relatively poor, this seems more related to the difficulty of assigning genres based on lyrics rather than a flaw in the model itself.

4.2 Comparison of Classification Model vs. Human Performance on the Generated Song Dataset

In this subsection, we analyze the performance of the classification model compared to human performance on the generated song dataset. As shown in

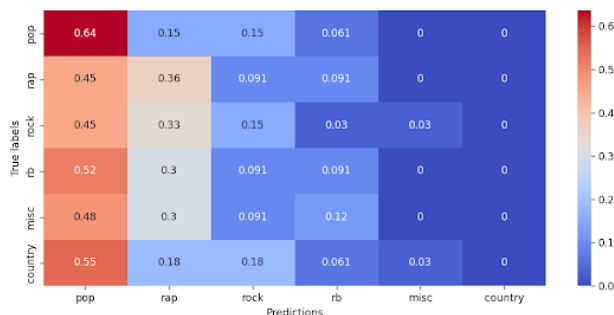


Figure 3: Heatmap of the classification model’s performance on a generated dataset of 198 songs, equally representing all six genres.

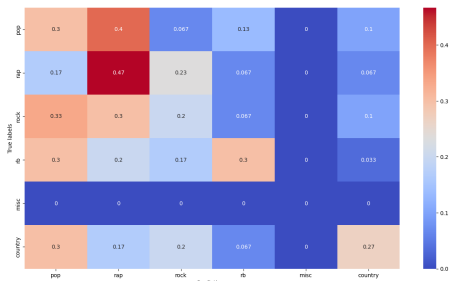


Figure 4: Aggregate human performance on genre classification for generated songs.

Figure 3, our classification model had difficulty accurately classifying the generated songs. Similar to the evaluation on the test dataset, the model significantly overpredicted the pop genre and exhibited the same tendency for rap. This issue appears to stem from the generation process, as we observe comparable results in the human baseline, as indicated in Figure 4. Although humans performed significantly better in some genres, such as country and R&B, their overall accuracy in classifying these songs remained low.

The model’s overall performance on generated songs reveals significant limitations. The low ac-

curacy in human annotations emphasizes the challenge of assigning genres based on generated lyrics. This finding suggests that either the generation process fails to capture genre-specific characteristics adequately, or the classifier struggles to generalize its learned patterns to synthetically created data. However due to the low performance of the human annotations, the former seems more likely. Future improvements could focus on enhancing the genre fidelity of generated lyrics by incorporating additional training constraints or fine-tuning the classifier on a dataset that includes generated examples.

4.3 Analysis of an Example-Generated Song

To further investigate the model’s lyric generation, we provide an example of a prompt the generation model received and the corresponding generated song. The song was generated to be classified under the country genre.

Prompt:

Prompt: Genre: Country Starting Lyric: "What they talkin' 'bout? They ain't talkin' 'bout nothin'" Output: What they talkin' 'bout? They ain't talkin' 'bout nothin'

Selected Lyrics for Analysis:

Below, we analyze a few individual lines from the generated song, focusing on their genre alignment and fluency.

Line 1: "Tell em where to get ya food boy (Food boy)"

Line 2: "Get outta our face don't act funny (Giggle)"

Line 3: "Well you always keep it poppin' some weed"

Line 4: "When you leave your crib some-one can hear you now you are back"

Analysis: In this particular example, we identify a key issue with the generation model. The classification model struggled significantly with categorizing country songs, as indicated in Figure 3. This raises an interesting question: Is the classification model failing due to its limitations, or is there a problem with the generation process?

The prompt instructed the model to produce a country song, but the provided starting lyric suggests a genre more aligned with rap or R&B. Analyzing the generated lyrics reveals that the model

created a song that closely resembles a rap piece, which is how it ultimately classified it. This situation indicates that there may be an issue with our prompting process, particularly with the inclusion of the starting lyric. It seems that the model is more inclined to continue the genre suggested by the starting lyric rather than adhering to the specified genre.

5 Discussion

The primary question of this study was whether a song's genre can be determined solely from its lyrics, and whether language models—specifically BERT for classification and GPT-2 for generation—can effectively analyze and replicate the relationship between lyrics and genre. This research is motivated by the understanding that musical genres often influence the thematic, cultural, and stylistic elements present in lyrics.

Our results showed that genre can be inferred from song lyrics with reasonable accuracy, both using the BERT classification model and by human annotators. When we compared the performance of the model with that of human classifiers, we found that the model performed well. This is supported by human annotators making more errors than the model, likely due to the subjective nature of genre classification. Furthermore, our analysis of lyrics generated by GPT-2 revealed challenges in ensuring that the generated songs matched the intended genre, with some songs failing to align correctly. These findings suggest that while lyrics serve as a strong indicator for genre classification, the relationship between genre and lyrics is complex and may require more nuanced approaches for both classification and generation tasks.

Several limitations in this study should be noted. First, the dataset used for both classification and generation was imbalanced, with some genres underrepresented. This imbalance may have affected the model's ability to classify and generate songs for less-represented genres. Furthermore, the genre classification model, while effective, could not always distinguish between genres with overlapping thematic elements, such as pop and rock. Additionally, the GPT-2 generation model sometimes struggled with genre alignment, particularly when the starting lyric was incongruent with the intended genre, which highlights potential limitations in the prompt engineering process.

Future research could address the issue of dataset

imbalance by creating a more balanced collection of songs, which would ensure better performance across all genres. Additionally, exploring more advanced generation models, such as LLAMA could improve adherence to specific genres in the generated lyrics. This study also opens the door to incorporating more sophisticated evaluation metrics that extend beyond classification accuracy. For example, measures of coherence, fluency, and thematic consistency in generated lyrics could provide deeper insights into the quality of generated lyrics. In conclusion, this study demonstrates that song lyrics can be an effective indicator of musical genre, both for classification purposes and for generating genre-specific lyrics. While language models such as BERT and GPT-2 show promise in these tasks, there are still challenges in achieving accurate genre classification and ensuring that the generated lyrics align with specific genres. The findings suggest that further research is needed to improve datasets, models, and evaluation methods. This will help refine our understanding of the interaction between genre and lyrics, as well as enhance the capabilities of natural language processing systems in this area.

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